

catalyst A substance that speeds up a chemical reaction without being changed itself at the end of the reaction.

cathode A negative *electrode*, from which electrons flow or are emitted.

cathode ray tube A kind of vacuum tube incorporating a fluorescent screen, best known from its use in televisions and monitors before the era of flat screens.

cauterization Destroying tissues by applying heat: used in medicine, especially in the past, to remove small growths or stop bleeding.

celestial body A natural body in space, such as a planet or star.

celestial sphere Imaginary sphere on which the stars seem to lie when seen from Earth.

cell (1) The "unit of life": a tiny structure composed of genes, surrounding fluid that carries out chemical reactions, and an enclosing membrane. See also *eukaryotic cell*, *prokaryotic cell*. (2) See *voltaic cell*.

cell division The process by which one cell splits to produce two daughter cells.

Celsius scale The temperature scale in which, under normal conditions, water freezes at 0° and boils at 100°.

centrifuge A device used to separate substances of different densities by spinning them at high speed.

cerebellum Part of the brain near the back of the skull. Its primary role is to control the detailed coordination of movements.

cerebrum The largest part of the brain in mammals, and in humans responsible for most conscious thought and activity.

chain reaction A chemical or nuclear reaction in which the product of one step triggers the next step, which in turn triggers the next step, and so on.

chaos theory A mathematical theory for analyzing complex systems whose behavior is very dependent on initial conditions, for example weather systems.

charged particle A small particle which has a net positive or negative electrical charge.

chlorophyll The green pigment found in plants that absorbs light to provide the energy for photosynthesis.

chloroplasts The chlorophyll-containing structures in the cells of plants and *algae* where photosynthesis occurs.

chromatid One of two identical strands of a chromosome. During cell division, the strands part and become separate *chromosomes*.

chromosomes Structures within living cells that contain copies of the genes of an organism. Each chromosome consists of a single long DNA molecule combined with various proteins. For example, humans have 23 pairs of chromosomes, with a complete set present in nearly every cell of the body.

circuit See *electric circuit*.

circulation See *vascular circulation*.

circumference The distance round the perimeter of an object or shape.

climate The average weather conditions of a region over a long period.

clone An identical copy or set of copies. Depending on the context, it can refer to: a copied DNA molecule; a set of identical

descendants of a given cell; an animal bred artificially using the cell nucleus of an adult.

cloud chamber An early form of apparatus for detecting subatomic particles.

codon A sequence of three adjacent *bases* that forms part of the genetic code. Most codons represent the code for adding a specific *amino acid* to a protein being synthesized in the cell.

cohesion The force of attraction between two particles of the same substance.

coke A solid fuel, mostly carbon, obtained by heating coal in the absence of air.

combustion A chemical reaction (burning) in which a substance combines with oxygen, producing heat energy.

comet Any of millions of bodies in the outer Solar System consisting of a mixture of rocky particles and ice. A comet becomes apparent when it orbits close to the Sun, its evaporating ice and dust particles producing a visible tail.

companion planting Growing plants of different crops together for mutual benefit.

compass Any of various devices to indicate the direction of north and south.

compound A *molecule* or chemical substance made up of atoms of two or more elements bonded together.

concave Curving or projecting inward.

concentric spheres Hollow spheres, arranged one outside the other, with the same center.

conductor A structure or material that conducts electricity and/or heat easily.

cones Light-sensitive cells in the retina of the eye of humans and some other animals that make it possible to see colors.

conic section Any of several mathematically important curves and shapes, produced by intersecting a cone with a plane surface.

conjugation In bacteria, the transfer of genetic material by direct cell-to-cell contact.

conjunctiva Mucous membrane covering the inside of the eyelids and the front of the eye.

conservation of energy The principle that energy can neither be created nor destroyed, but only changed from one form to another.

constellation A named pattern of stars that astronomers use when referring to different regions of the sky.

continental drift The moving of Earth's continents with respect to each other over millions of years, resulting from plate tectonic activity.

convection The transfer of heat through a fluid by currents within the fluid.

convergent boundary The line along which two tectonic plates that are moving toward each other meet. See *plate tectonics*.

convergent evolution The phenomenon by which unrelated species evolve similar features as a result of *adaptation* to similar environments or ecological niches.

convex Curving or projecting outward.

Coriolis effect The deflection of winds and ocean currents by the rotation of Earth.

corona The outer atmosphere of the Sun or another star.

cosine (cos) The ratio of the adjacent side of an angle to the hypotenuse in a right-angled triangle. Also, the mathematical function

describing how this ratio varies when the angle changes as the hypotenuse sweeps round a circle.

cosmic background radiation Microwave radiation coming from all directions of outer space that represents a relic of the Big Bang.

cosmic rays High-energy particles bombarding Earth from space.

cosmological principle The principle that the Solar System and Earth are not positioned in any special or central place in the Universe.

cosmology The study of the Universe on the largest scale.

coulomb The SI unit of electric charge.

covalent bond A chemical bond formed by atoms sharing one or more electrons.

cross-fertilization Fertilization of a plant by a different member of the same species (in contrast to self-fertilization).

crystal A solid whose constituent atoms, ions, or molecules are arranged in a regular, repeating geometrical pattern.

cubic equation A mathematical equation containing at least one variable number multiplied by itself twice (for example, $x \times x \times x$, also written x^3), but no variables multiplied more times than this.

cumulus clouds Rounded fluffy clouds that form when moisture-containing air rises.

cuneiform A form of writing using wedge-shaped impressions made in clay, characteristic of some ancient civilizations.

curvature of space The idea derived from relativity that, viewed on a large scale, space itself is curved, not in three straight dimensions as common sense would suggest.

curve In mathematics, a plot on a graph of one quantity against another, or a line that represents a particular geometrical shape.

dark energy A little-understood theoretical phenomenon proposed to explain why the expansion of the Universe is accelerating.

dark matter Matter that cannot be detected by conventional means but which must exist in galaxies to explain their gravitational properties. Its properties mean that it cannot be made of atoms as we understand them.

dead reckoning Navigating by estimating speed and direction only, without using other checks such as astronomical observations.

decibel The standard unit for measuring sound intensity.

decomposition (1) The decay of organic material. (2) In chemistry, a reaction that breaks a larger *molecule* into smaller ones.

detector In electronics, the circuit in a radio receiver that separates out the sound signal from a radio wave.

differential calculus See *calculus*.

differentiation The type of calculation carried out in differential calculus.

diffraction The bending of waves around obstacles or the spreading out of waves when they pass through a narrow aperture.

diffusion The spreading of one substance through another by random motions of its atoms or molecules.

digital A term referring to the storage and transmission of information (such as sound or video information) using patterns of

discrete units, such as the 0 and 1 values of the binary system.

digital sound Sound recorded digitally.

diode An electronic component that lets electricity flow in one direction only.

diopetre A unit of refractive power of a lens. See *refraction*.

diploid cell A cell containing two copies of each chromosome.

DNA Short for deoxyribonucleic acid, the large *molecule* that carries the genetic information in all living things, except for some viruses which use *RNA*.

double helix A double spiral: the term is used particularly to refer to the two intertwining strands of a DNA molecule.

driving mechanism A mechanism that transmits mechanical movement and power.

dwarf planet An astronomical object (which includes the former planet Pluto) that is big enough to have become rounded by its own gravity, but not big enough to have cleared surrounding space of objects.

dye A substance that colors a material.

dynamics Branch of physics, which studies the movement of objects when under the influence of forces.

dynamo A generator that produces direct current.

eclipse The temporary hiding of one astronomical object behind another, especially the Sun behind the Moon when seen from Earth (a solar eclipse), or Earth between the Moon and the Sun, when Earth's shadow falls on the Moon (a lunar eclipse).

ecliptic The curved path, representing the plane of the Solar System, through which the Sun and planets appear to move through the skies over the course of a year.

ecology The study of the relationships between organisms and their environment.

ecosystem A community of living things considered together with the interactions between them and their physical environment.

egg (1) A female sex cell (gamete), also called an ovum. (2) A structure that protects the growing embryo in birds and other animals.

elasticity The tendency of a substance to "bounce back" to its original shape or volume when an applied force is removed.

electric charge A basic property of many subatomic particles that makes them interact electromagnetically. Charge can be either positive or negative.

electric circuit A complete loop of conducting material that carries an electric current and connects electrical devices such as switches and light bulbs.

electric current The flow of electrical energy.

electric motor A device in which electrical power is converted into rotational mechanical power.

electrical resistance Resistance to the flow of electricity, usually resulting in heat being given off.

electrode An electrical terminal, which can conduct electricity into or out of a system.

electrolysis Chemical change or breakdown in an electrolyte caused by electrical current being passed through it.